COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	08/13/2026	**DATE SUBMITTED:**	08/13/2026
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PRELIM EXAM	**Installing Ansible, Building an Inventory, Using Metagroups, Variables, and Creating a Playbook**

1. PROCEDURE OUTPUT

HOSTNAME	Operating System	IP Address
Server 1	Ubuntu	192.168.199.152
Server 2	Ubuntu	192.168.199.6
Workstation	Ubuntu	192.168.199.153
Cent OS(get corrupted in the pc)		

Part A: Installing and Verifying Ansible
Submit a screenshot of the following:
- **Ansible version**
- **SSH connectivity**

```
charles-vence-espinas@workstation:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/charles-vence-espinas/.ansible/plugins
/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/charles-vence-espinas/.ansible/collections
:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/p
ython3)
  jinja version = 3.1.2
  libyaml = True
```

```
charles-vence-espinas@workstation:~$ ssh charles-vence-espinas@192.168.199.152
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Last login: Thu Aug 13 08:49:19 2026 from 192.168.199.153
charles-vence-espinas@server1:~$
logout
Connection to 192.168.199.152 closed.
```

```
charles-vence-espinas@workstation:~$ ssh charles-vence-espinas@192.168.199.6
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Last login: Thu Aug 13 08:49:04 2026 from 192.168.199.153
charles-vence-espinas@server2:~$
logout
Connection to 192.168.199.6 closed.
```

Part B: Building an Inventory File

Demonstrate the following:

- inventory.yaml
- Successful inventory verification
- Successful ping results

charles-vence-espinas@workstation: ~/ansible-lab

GNU nano 7.2 inventory.yaml

```
all:
  children:

    ubuntu:
          hosts:
            server1:
              ansible_host: charles-vence-espinas@192.168.199.152

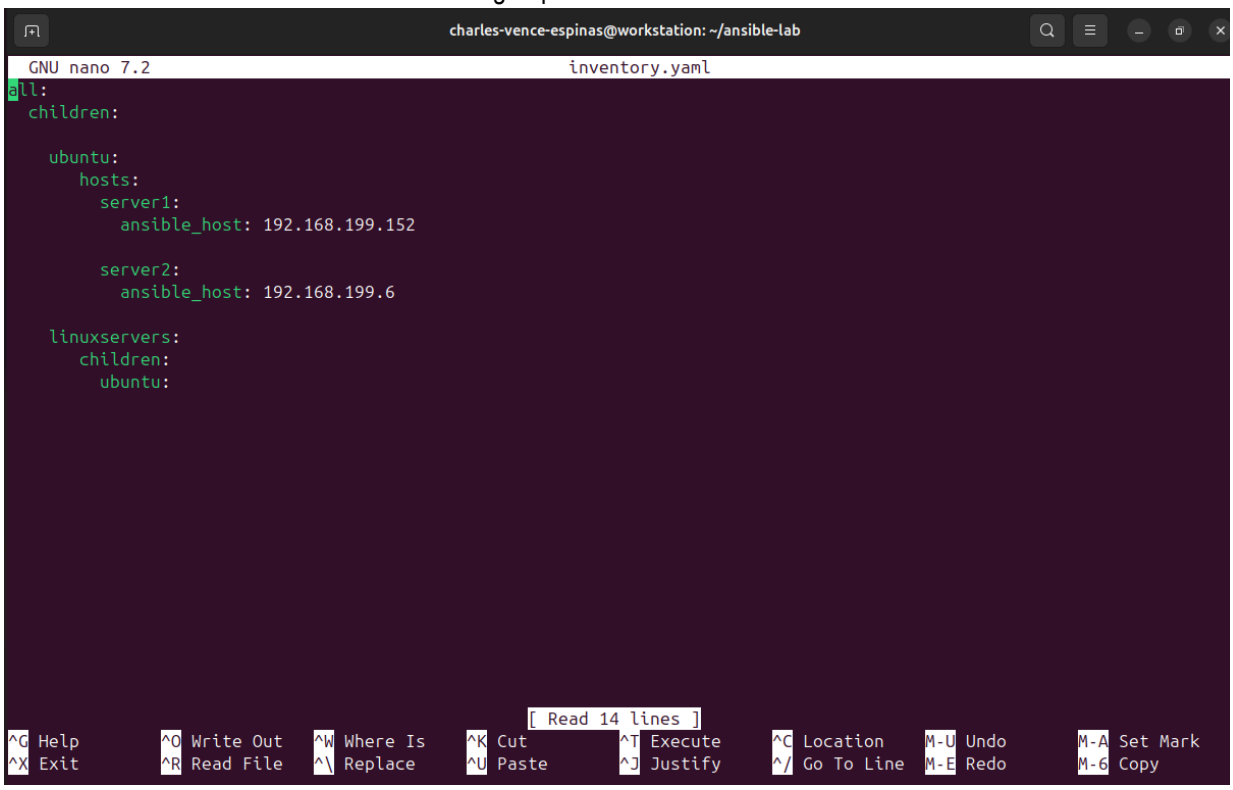
            server2:
              ansible_host: charles-vence-espinas@192.168.199.6
```

```
charles-vence-espinas@workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
    "_meta": {
        "hostvars": {
            "server1": {
                "ansible_host": "charles-vence-espinas@192.168.199.152"
            },
            "server2": {
                "ansible_host": "charles-vence-espinas@192.168.199.6"
            }
        }
    },
    "all": {
        "children": [
            "ungrouped",
            "ubuntu"
        ]
    },
    "ubuntu": {
        "hosts": [
            "server1",
            "server2"
        ]
    }
}
charles-vence-espinas@workstation:~/ansible-lab$ ansible all --list-hosts
  hosts (2):
    192.168.199.152
    192.168.199.6
```

```
charles-vence-espinas@workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Part C: Creating a Metagroup

Demonstrate successful execution of the metagroup



```
charles-vence-espinas@workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

```
charles-vence-espinas@workstation:~/ansible-lab$ ansible linuxservers --list-hosts -i inventory.yaml
  hosts (2):
    server1
    server2
```

Part D: Creating Variables

Demonstrate successful use of variables.

charles-vence-espinas@workstation: ~/ansible-lab

GNU nano 7.2 inventory.yaml

```
all:
  children:

    ubuntu:
       hosts:
         server1:
           ansible_host: 192.168.199.152
           os_type: Ubuntu

         server2:
           ansible_host: 192.168.199.6
           os_type: Ubuntu

    linuxservers:
       children:
         ubuntu:
          vars:
            ansible_user: charles-vence-espinas
```

```
charles-vence-espinas@workstation: ~/ansible-lab
 GNU nano 7.2                                  testvars.yaml
---
- name: Display Variables
  hosts: linuxservers

  tasks:

    - name: Show Operating System
      debug:
        msg: "Host {{ inventory_hostname }} is running {{ os_type }}"
```

```
charles-vence-espinas@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvars.yaml

PLAY [Display Variables] ***************************************************************************************************

TASK [Gathering Facts] *****************************************************************************************************
ok: [server1]
ok: [server2]

TASK [Show Operating System] ***********************************************************************************************
ok: [server1] => {
    "msg": "Host server1 is running Ubuntu"
}
ok: [server2] => {
    "msg": "Host server2 is running Ubuntu"
}

PLAY RECAP *****************************************************************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Part E: Creating Your First Playbook

Demonstrate:

- Successful playbook execution
- Successful task completion on all managed nodes

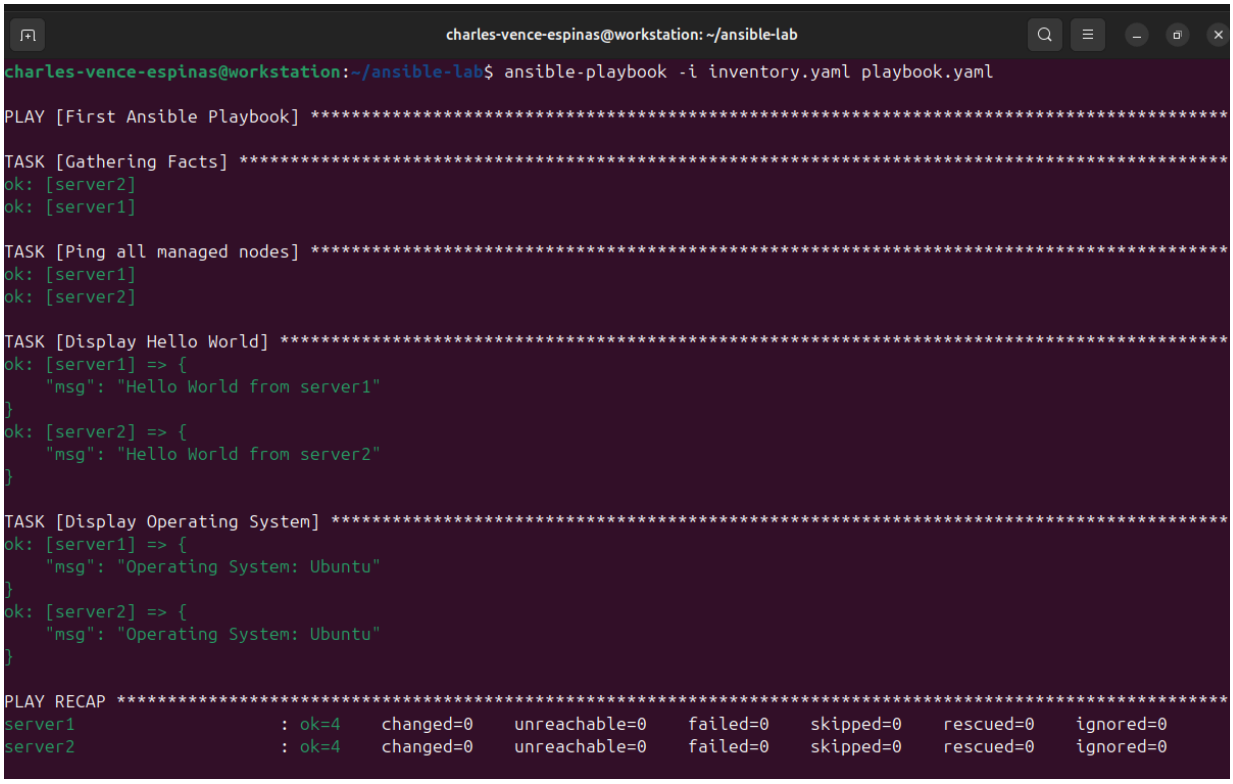
```
charles-vence-espinas@workstation: ~/ansible-lab
 GNU nano 7.2                                  playbook.yaml
---
- name: First Ansible Playbook
  hosts: linuxservers

  tasks:

      - name: Ping all managed nodes
        ping:

      - name: Display Hello World
        debug:
          msg: "Hello World from {{ inventory_hostname }}"

      - name: Display Operating System
        debug:
          msg: "Operating System: {{ os_type }}"
```



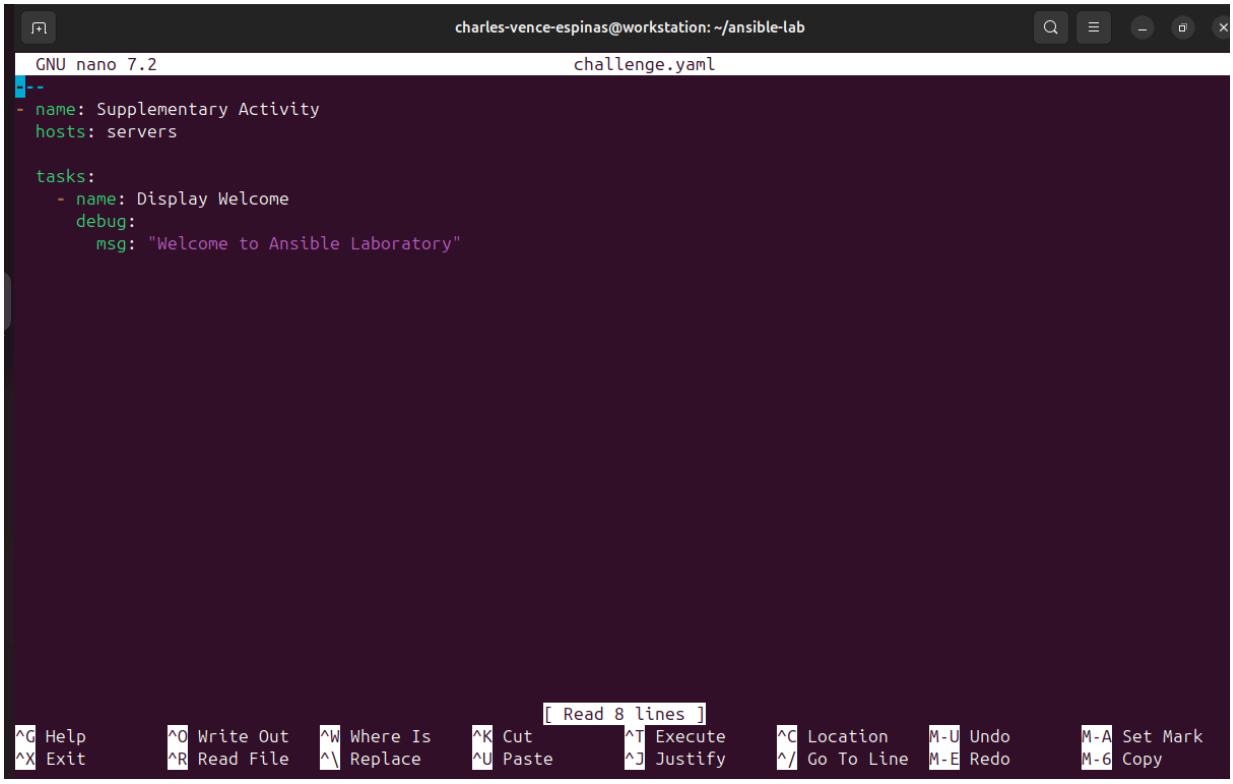
2. ANSWERS TO SUPPLEMENTAL ACTIVITY

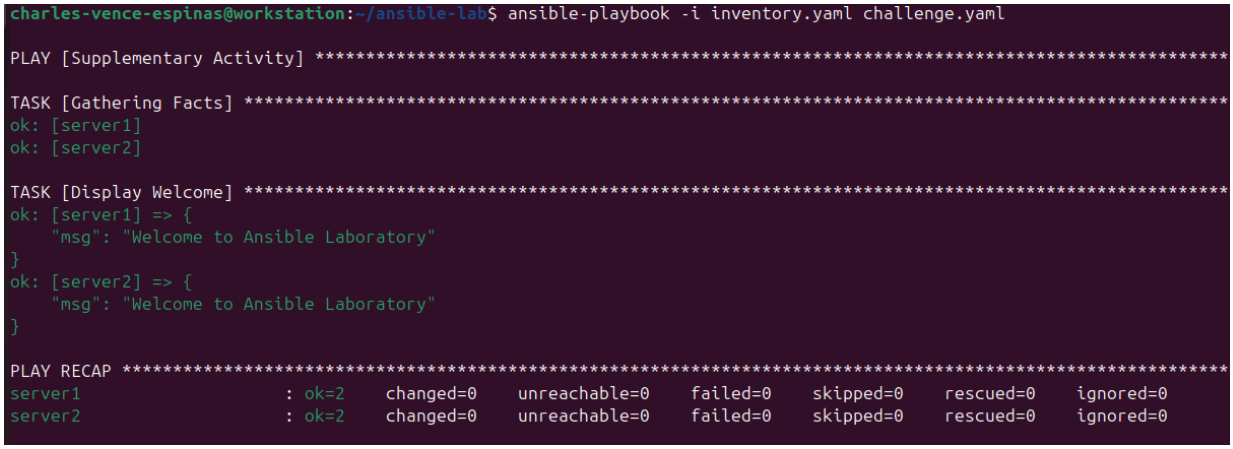
Create a new group named servers that contains only:

- ubuntu1
- centos1

Create a playbook named challenge.yaml that displays the message

Run the playbook and demonstrate that only hosts in the servers group receive the message.





Guide Questions

1. What is the role of the Control Node in Ansible?

- The control node is the one that acts as a central command center in the ansible. It is the main virtual machine in which we write and run the commands or playbooks to manage and configure all the remote servers.

2. Why is an inventory file required?

- Inventory file is needed because it acts as ansible address book and without it, ansible will not know which server exist, what their ip addresses or hostnames are and how they are categorized into different groups.

3. What is the purpose of a metagroup?

- The metagroup is a group of groups and it is defined using children and the purpose of it is to combine many smaller groups into one larger master group. This allows us to run a single ansible command or playbook against many different server groups at the same time instead of targeting them one by one.

4. How do variables simplify Ansible administration?

- Variables simplify ansible administration by replacing the details like usernames, IP addresses or port numbers with flexible placeholders. Instead of changing a value in many places, we can update it once in a variable file in which makes the playbooks easier to reuse and maintain.

5. What is the difference between an ad hoc command and a playbook?

- An ad hoc command is a quick one line command used for instant tasks while a playbook is a saved file containing a full list of automated tasks.

Conclusion

- In conclusion, I learned that using Ansible to manage control nodes and managed nodes is a verygood and effective way to control multiple servers from a single central location. It simplifies the system administration by replacing repetitive manual tasks with structured inventory files, variables, and automated playbooks. Also, this approach saves time, reduces human error, and many more that it makes it straightforward and reliable.

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Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

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